Architectural Capability Review: Consolidating Vehicular Edge Gateway Functions into ACME oneM2M CSE

Feasibility Study, Application Entity Multi-Hazard Analysis, and Technical Roadmap for ETSI TS 118 101 Middleware

DOCUMENT ID	RELEASE / STATUS	STANDARD COMPLIANCE	TARGET HARDWARE
ARC-SDV-CSE-2026-01	Version 2.0.0 • Approved	ETSI TS 118 101 / oneM2M TS-0010	Raspberry Pi 5 / ACME IN-CSE

Executive Summary & Comprehensive Scope Formulation

In the current Software-Defined Vehicle (SDV) baseline, the Vehicular Multi-Access Edge Computing (MEC) Gateway executes a seven-tier microservice mesh (Mosquitto, Node-RED, ACME CSE, Bridge AE, InfluxDB, Grafana, Portal) consuming ~650 MB RAM. While initial discussions centered narrowly on thermal divergence (ΔT) and blower fan PWM, an audit of the formal ETSI TS 118 101 resource tree, the Insomnia fleet harness, and hardware schematics demonstrates that **the Application Entity (AE) encompasses 7 standardised containers, a 4-channel LED actuator matrix, tachometer feedback, multi-hazard health classification (CO safety, humidity drift ΔRH, fan stall, and exterior-cabin AQI differential ΔIAQ), and dynamic fleet group fan-out**. ACME IN-CSE natively possesses the capacity to absorb ingress bridging, rule evaluation, edge buffering, and fleet fan-out, reducing gateway memory by 61.5% and closed-loop latency by up to 60%.

1. THE COMPREHENSIVE SCOPE OF THE VEHICLE APPLICATION ENTITY (AE)

The vehicle Application Entity (`m2m:ae`, e.g. `vehicle-001..N`) represents a complete automotive-grade cabin life-support and environmental regulation subsystem:

AE SEMANTIC CONTAINER	TRANSDUCER INTERFACE & BUS	ENVIRONMENTAL CONCERN & CONTROL ACTUATION
`cnt_temperature-cabin`	Dual SCD30 (I²C Fast-Mode) & DHT22 (1-Wire)	Cabin thermal comfort; triggers Auxiliary Heater Core (Red LED DO) when T < 19.0 °C.
`cnt_humidity-cabin`	Dual SCD30 & DHT22 relative humidity (%)	Window defogging; triggers Dehumidifier / Alert (Amber LED DO) when RH > 65%.
`cnt_co2-cabin`	Sensirion SCD30 optical NDIR (ppm)	Passenger alertness; scales Purge Blower (Blue LED PWM, 25 kHz) when CO_2 > 800 ppm.
`cnt_co-cabin`	Carbon Monoxide toxic gas sensor (ppm)	Life-safety hazard; wired to `sub_co_alert` dispatching immediate emergency alarm.
`cnt_air-quality-cabin`	VOC / Particulate IAQ sensor (0–500)	Cabin indoor air quality index; detects organic pollutants and smoke.
`cnt_air-quality-exterior`	Exterior ambient air quality sensor (AQI)	Governs Fresh Air Intake vs Cabin Air Recirculation Flap damper based on ΔIAQ.
`cnt_raw-telemetry`	Unified multi-sensor frame & metrics	Synchronised frame with GNSS, timestamps, PWM duty cycle, and Tachometer RPM.

Physical Actuator Array: Purge Fan (Blue LED PWM, GPIO 18, 0–100%), Auxiliary Heater (Red LED DO, GPIO 17), Eco-Vent (Green LED DO, GPIO 16), Alert/Dehumidifier (Amber LED DO, GPIO 15), Optical Node Annunciator (WS2812B RGB, GPIO 38), and Hall-Effect Commutation Tachometer (GPIO 19, PCNT Unit 0).

2. MULTI-DIMENSIONAL DATA HEALTH CLASSIFICATION IN `BRIDGE_AE`

In a production cyber-physical automotive architecture, data integrity cannot rely solely on temperature delta and network latency. The Data Health Engine must evaluate five interconnected mathematical criteria:

```
1. Latency Freshness:     τ_age = t_eval - t_sample ≤ 2.5 s (STALE if τ > 2.5 s, FAULT if τ > 10.0 s)
2. Thermal Divergence:    ΔT = |T_SCD30 - T_DHT22| ≤ 1.5 °C (DEGRADED if ΔT > 1.5 °C)
3. Humidity Divergence:   ΔRH = |RH_SCD30 - RH_DHT22| ≤ 10.0% (DEGRADED if ΔRH > 10%)
4. Actuator Stall Fault:  (PWM ≥ 30%) AND (Tachometer RPM < 100) ⇒ FAULT (Blower Jam / Line Break)
5. Toxic Gas Hazard:      CO ≥ 30.0 ppm ⇒ FAULT / CRITICAL EMERGENCY (Immediate Purge Preemption)
6. Damper Flap Control:   ΔIAQ = IAQ_exterior - IAQ_cabin ⇒ Drives Fresh Air vs Recirculation Damper
```

HEALTH STATE	DETERMINISTIC EVALUATION CONDITIONS	OPERATIONAL HVAC & ACTUATION RESPONSE
FRESH	τ_age ≤ 2.5s • ΔT ≤ 1.5°C • ΔRH ≤ 10% • Tachometer Valid	Nominal closed-loop regulation. WS2812B illuminates Green.
STALE	τ_age ∈ (2.5s, 10.0s] • Network jitter or cellular handover	Hold last verified setpoint; warn supervisory cockpit. WS2812B Amber.
DEGRADED	ΔT > 1.5°C OR ΔRH > 10% (transducer membrane drift/fouling)	Fail-safe conservative scaling; Amber Alert LED active. WS2812B Amber.
FAULT	τ_age > 10s OR Null Reading OR Blower Motor Stall OR CO ≥ 30ppm	Autonomous hardware override (50% purge); Red LED / Alarm. WS2812B Red.

3. EDGE GATEWAY VS ACME CSE COMPARATIVE CAPABILITY MATRIX

GATEWAY SUBSYSTEM	CURRENT GATEWAY IMPLEMENTATION	ACME CSE NATIVE MECHANISM	FEASIBILITY & IMPACT
MQTT Ingress / Egress Bridging	Node-RED IPE container translating MQTT `sdv/+/telemetry` to HTTP POST `<cin>` and downlink webhooks	**oneM2M TS-0010 Native MQTT Protocol Binding** (`[mqtt]` binding in `acme.ini`)	HIGH FEASIBILITY Eliminates Node-RED completely. Reclaims ~200 MB RAM.
Multi-Hazard Health Classification	Python Flask `bridge_ae` evaluating latency, ΔT, ΔRH, stall, and CO	**ACME CSE Scripting Engine** (`[scripting]` / `onCreate` hooks) or `<flexContainer>` rules	HIGH FEASIBILITY Internalises all 5 health checks within the CSE.
Edge Time-Series Retention	InfluxDB 2.x columnar database with 30-day retention policies	**oneM2M `<timeSeries>` & Container Retention** (`mni`, `mbs`, `mia` TTL expiry)	MEDIUM FEASIBILITY Native edge buffering & missing data detection.
Downlink Actuator Routing	Node-RED `/downlink` webhook receiver converting `<sgn>` to MQTT publish	**Native oneM2M Subscription to MQTT Topic** (`nu: ["mqtt://..."]` or Request Topics)	HIGH FEASIBILITY Direct notification dispatch without HTTP proxy hops.
Access Control & Multi-Tenancy	Static Mosquitto `acl.conf` text file matching client CN	**oneM2M Access Control Policy (`<acp>`)** with Originator CRUDND rules	HIGH FEASIBILITY Dynamic semantic policy governance per AE-ID.
Fleet Group Operations & Fan-Out	Custom application iteration loops & multi-request scripts	**oneM2M Group Entity (`<grp>`) with FanOutPoint (`<fopt>`)**	FULLY PROVEN Atomic fleet queries (`/fopt/la`) & broadcast.
Administrative Observability	Custom Python Flask `web_portal` container on port 8000	**ACME Built-in Web UI & Management Endpoint** (`[webui]` in `acme.ini`)	HIGH FEASIBILITY Live interactive 7-container tree navigation & logs.

4. DETAILED TECHNICAL AUDIT: SPECIFIC CAPABILITIES ACME CSE CAN ABSORB

4.1 Ingress & Egress Bridging via oneM2M TS-0010 MQTT Binding

Currently, Node-RED acts as an Interworking Proxy Entity (IPE) bridging raw MQTT messages to oneM2M HTTP primitives. By enabling `[mqtt]` in `acme.ini`, ACME CSE establishes an active MQTT client connection directly to Mosquitto, subscribing to standard oneM2M request topics:

```
Uplink Telemetry Ingestion:
  PUBLISH /oneM2M/req/C_CabinNode_Car01/id-sdv/json
Downlink Directives (Direct Notification):
  SUBSCRIBE /oneM2M/req/id-sdv/C_CabinNode_Car01/json
```

The ESP32-S3 firmware can publish ContentInstance creation primitives directly over MQTT, completely bypassing Node-RED. Actuation notifications are published by ACME directly back onto the vehicle's downlink topic. This removes the Node-RED container, reclaims ~200 MB RAM, and cuts closed-loop latency by 40–80 ms.

4.2 In-CSE Multi-Hazard Health Classification via ACME Scripts

ACME CSE's internal Turing-complete scripting engine (`acme-scripts`) allows registering an `onCreate` trigger against `cnt_raw_telemetry`. The script parses the incoming JSON, evaluates freshness ($\tau_{age} \leq 2.5$s), thermal divergence ($\Delta T \leq 1.5$°C), humidity divergence ($\Delta RH \leq 10\%$), motor stall (PWM ≥ 30% with RPM < 100), and CO toxic gas hazard, automatically tagging the <cin> with semantic labels (e.g. `lbl: ["health:FRESH", "delta_t:0.32", "delta_rh:1.8", "stall:false"]`) while instantly instantiating an alarm <cin> in the event of an emergency.

4.3 Native Time-Series Buffering, Storage Bounds & Gap Detection

In standard oneM2M TS-0001 deployments, historical sensor readings are maintained natively via <timeSeries> (<ts>) and <timeSeriesInstance> (<tsi>) resources. For standard containers, ACME natively enforces robust retention policies:

```
"m2m:cnt": {
  "rn": "raw-telemetry",
  "mni": 1440,      /* Max 1,440 instances (24h buffer at 1-min sampling) */
  "mbs": 2097152,   /* Strict 2 MB byte-size limit protecting flash storage */
  "mia": 86400      /* Max instance age: 24h TTL automatic pruning */
}
```

Furthermore, oneM2M <ts> provides built-in missing data detection (`missingDataDetect`), notifying upstream applications when sensor updates drop out without custom watchdog code.

4.4 Atomic Fleet Group Operations and Fan-Out (<grp> & <fopt>)

Demonstrated in Folder `05_Fleet_Registry_and_Discovery` of the Insomnia collection, ACME CSE natively handles fleet-wide fan-out aggregation. Rather than executing sequential queries across hundreds of vehicles, a single request to the FanOutPoint retrieves the latest synchronised snapshot across the entire fleet:

```
GET http://192.168.122.91:8080/cse-in/grp_fleet_telemetry/fopt/la
```

Simultaneously, broadcasting setpoints (e.g. emergency ventilation purge directives) to all vehicles is achieved with an atomic POST `.../grp_fleet_telemetry/fopt`.

4.5 Centralised Semantic Access Control Policies (<acp>)

Instead of maintaining brittle text-based broker ACLs (`acl.conf`), ACME CSE enforces standard <acp> resources (e.g. `acp_fleet_manager`), mapping CRUDND rights to Originator IDs (`CAdmin`, `C_vehicle_nodes`, `S*`). This centralises security governance within the data layer.

5. ARCHITECTURAL BOUNDARIES: WHAT MUST REMAIN DEDICATED

To preserve cryptographic isolation, high-frequency analytical performance, and cyber-physical safety, specific responsibilities must strictly remain separate from ACME CSE:

• Retained on Dedicated Edge Platform

- **Mutual TLS 1.3 Termination (Mosquitto):** Hardware-accelerated ECDSA P-256 handshakes, X.509 client verification, and port 8883 network isolation.
- **Long-Term High-Rate TSDB Analytics (InfluxDB):** Column-oriented storage, Flux math, downsampling, and continuous multi-week fleet metrics.
- **Operational Telemetry Cockpit (Grafana):** High-cardinality multi-series interactive dashboards and operator alerting.
- **Autonomous Hardware Fail-Safes (ESP32-S3):** FreeRTOS Core 1 closed-loop fan purge overrides executing deterministically during network blackouts.

• Consolidated into ACME oneM2M CSE

- **oneM2M Ingress & Egress Ingestion:** Native TS-0010 MQTT binding eliminating Node-RED translation proxies.
- **Short-to-Medium Telemetry Buffering:** Container retention policies (`mni`, `mbs`, `mia`) with automatic TTL pruning.
- **Empirical Multi-Hazard Health Classification:** Event-driven ACME scripts tagging instances and evaluating ΔT, ΔRH, stall, and CO.
- **Group Aggregation & Multicast Fan-Out:** Standard <grp> and <fopt> fan-out endpoints.
- **Fine-Grained Semantic Authorisation:** Dynamic oneM2M <acp> policies replacing static broker ACL files.

6. SYNCHRONISATION SPECIFICATION FOR ARCHITECTURE_DETAILED.PNG

To align the high-level diagram with the semantic tree and circuit schematic, the following updates are recommended:

DIAGRAM VISUAL BLOCK	RECOMMENDED SYNCHRONISATION ENHANCEMENTS
In-Cabin Sensing Node (ESP32-S3)	• **Sensors:** Add Carbon Monoxide (CO sensor), Cabin IAQ (VOC/PM), and Exterior Ambient AQI. • **Actuators:** Show all 4 LED channels: Blue (Purge PWM, GPIO 18), Red (Heater Core DO, GPIO 17), Green (Eco-Vent DO, GPIO 16), Amber (Alert DO, GPIO 15), and WS2812B Annunciator (GPIO 38). • **Feedback:** Illustrate Hall-effect tachometer feedback line from blower rotor into GPIO 19.
oneM2M Middleware Tier (ACME IN-CSE)	• Replace generic `/sdv-cse` container box with explicit **7-container vehicle AE subtree**: `[temp, hum, co2, co, iaq-cab, iaq-ext, raw-telemetry]`. • Add **Fleet Governance Entities**: <acp> (`acp_fleet_manager`) and <grp> (`grp_fleet_telemetry` with <fopt>). • Illustrate multi-vehicle nodes (`vehicle-001`, `vehicle-002`, `vehicle-N`) fanning into the central CSE.
Subscription Ingestion Routes	• Differentiate three distinct flows: (1) `sub_downlink` to Mosquitto, (2) `sub_co_alert` to Emergency Monitor webhook, and (3) `sub_fleet_telemetry` to Bridge AE / Cloud Twin.
Bridge AE Health Classifier Box	• Annotate all 5 multi-hazard checks: Freshness ($\tau_{age} \leq 2.5$s), Thermal Delta ($\Delta T \leq 1.5$°C), Humidity Delta ($\Delta RH \leq 10\%$), Blower Stall Fault Check, and Toxic CO Safety Guard.
Grafana Operations Cockpit	• Add Fleet Vehicle Selector (`vehicle-001..N`), Multi-Actuator status indicators (Heater, Eco, Purge, Dehumidifier), Tachometer RPM gauge with Stall Warning, and Emergency Toxic CO Banner.

7. STRATEGIC PHASED IMPLEMENTATION ROADMAP

PHASE	ACTION ITEMS & CONFIGURATION CHANGES	AFFECTED SERVICES / COMPONENTS	DELIVERABLE & GAIN
Phase 1: Quick Wins *Immediate*	1. Enable ACME native Web UI in `acme.ini` (`[webui] enable = true`). 2. Enforce 24-hour TTL expiry (`mia: 86400`, `mni: 1440`) across container manifests in the SOP.	`sdv_onem2m_cse` (ACME), SOP container provisioning manifest	Immediate tree visibility on gateway; automatic memory and flash garbage collection.
Phase 2: Protocol Consolidation *Target: Month 2*	1. Activate `[mqtt]` binding in `acme.ini` connecting directly to Mosquitto. 2. Upgrade ESP32-S3 firmware to emit oneM2M MQTT primitives directly to `/oneM2M/req/...`. 3. Decommission `nodered_ipe` container in `docker-compose.yml`.	`firmware/esp32s3_cabin_node`, `nodered_ipe`, `sdv_mosquitto_broker`	Saves ~200 MB RAM; eliminates intermediate HTTP translation; reduces latency by 40–80 ms.
Phase 3: Logic Consolidation *Target: Month 3*	1. Port multi-hazard classifier (ΔT, ΔRH, stall, CO) into an ACME internal script. 2. Configure direct HTTP notification from ACME CSE to InfluxDB (or lightweight Telegraf forwarder). 3. Decommission `bridge_ae` container.	`sdv_bridge_ae`, ACME script repository, `sdv_influxdb`	Shrinks edge mesh from 7 to 4 containers; health metadata directly embedded in semantic tree.

8. QUANTIFIED EDGE RESOURCE FOOTPRINT IMPACT ASSESSMENT

SYSTEM PERFORMANCE METRIC	BASELINE 7-CONTAINER MESH	STREAMLINED 4-CONTAINER MESH	NET OPERATIONAL GAIN
Total Gateway Memory Consumption (RAM)	~650 MB LPDDR4X	~250 MB LPDDR4X	**61.5% RAM Reduction (~400 MB saved)**
Active Edge Microservices / Daemons	7 containers (Mosquitto, Node-RED, ACME, Bridge, Influx, Grafana, Portal)	4 containers (Mosquitto, ACME CSE, InfluxDB, Grafana)	**42.8% Fewer Daemons to Maintain**
Inter-Container Network Transits per Sample	5 hops (ESP → Mosquitto → Node-RED → ACME → Bridge → Influx)	3 hops (ESP → Mosquitto → ACME → Influx)	**40% Fewer Intermediate Network Hops**
Closed-Loop Actuation Latency (τ_{prop})	120 – 280 ms	45 – 90 ms	**Up to 60% Closed-Loop Latency Reduction**
Configuration Drift & Security Surface	High (Separate Mosquitto ACLs, Node-RED flows, Python environment, ACME config)	Low (Single semantic source of truth governed by oneM2M Access Control Policies)	**Unified ETSI TS 118 101 Compliance**

9. ENGINEERING CONCLUSION & SIGN-OFF

This review concludes that ACME oneM2M IN-CSE is mature, feature-rich, and capable of replacing the Node-RED ingress proxy, Python multi-hazard health classification daemon, and edge web portal without sacrificing data integrity. Preserving Eclipse Mosquitto for hardware mTLS 1.3 termination and InfluxDB/Grafana for multi-series analytics strikes the optimal balance between **strict ETSI TS 118 101 semantic compliance**, **cyber-physical safety**, and **extreme edge resource efficiency**.

Lead Architecture Review:
Distributed Cyber-Physical Systems (FHTW)

Middleware Verification:
ACME oneM2M IN-CSE Core (Group MIO3B)

Document Authentication:
FHTW-AIOT-SDV-2026-PDF-02